Exponential Functions & Growth Models

ANSWER KEY

Section 1: Exponential Growth and Decay Fundamentals

1 Which function represents exponential decay with an initial value of 450 and a decay rate of 8% per year?

a. $f(t) = 450(0.92)^t$ ✓

b. $f(t) = 450(1.08)^t$

c. $f(t) = 450(0.08)^t$

d. $f(t) = 450 - 0.08t$

In exponential decay, the base is $b = 1 - r$. With $r = 0.08$, $b = 1 - 0.08 = 0.92$. The model is $f(t) = 450(0.92)^t$.

2 An exponential function passes through $(0, 6)$ and $(2, 54)$. If $f(x) = ab^x$ where $b > 0$, what is the value of b?

a. 3 ✓

b. 9

c. 6

d. $\sqrt{3}$

From $(0, 6)$, $a = 6$. From $(2, 54)$, $6b^2 = 54 \Rightarrow b^2 = 9 \Rightarrow b = 3$ (since $b > 0$).

3 If a quantity grows continuously at an annual rate of 5%, what is the continuous growth model for an initial investment of 1200 dollars?

a. $A(t) = 1200e^{0.05t}$ ✓

b. $A(t) = 1200(1.05)^t$

c. $A(t) = 1200(0.05)^t$

d. $A(t) = 1200e^{1.05t}$

Continuous compounding or continuous growth is modeled by $A(t) = Pe^{rt}$. Here $P = 1200$ and $r = 0.05$, yielding $A(t) = 1200e^{0.05t}$.

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Section 2: Doubling Time & Half-Life Calculations

4 A bacterial culture doubles every 4 hours. If there are initially 500 bacteria, which equation models the population after t hours?

a. $P(t) = 500 \cdot 2^{t/4}$ ✓

b. $P(t) = 500 \cdot 2^{4t}$

c. $P(t) = 500 \cdot 4^{t/2}$

d. $P(t) = 500 + 2^{t/4}$

The doubling model is $P(t) = P_0 \cdot 2^{t/d}$, where d is the doubling time. Here $d = 4$, so $P(t) = 500 \cdot 2^{t/4}$.

5 A radioactive isotope has a half-life of 15 days. If a sample contains 80 grams initially, how much remains after 45 days?

a. 10 g ✓

b. 20 g

c. 5 g

d. 15 g

The number of elapsed half-lives is $\frac{45}{15} = 3$. Remaining amount: $80 \cdot \left(\frac{1}{2}\right)^3 = 80 \cdot \frac{1}{8} = 10$ g.

6 If the doubling time of an investment is 12 years, what is the annual growth factor b in $A(t) = A_0 b^t$?

a. $2^{1/12}$ ✓

b. $12^{1/2}$

c. 2×12

d. $\frac{2}{12}$

Doubling time satisfies $b^{12} = 2$. Taking the 12th root yields $b = 2^{1/12} \approx 1.0595$.

Section 3: Equivalent Exponential Forms & Conversions

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7 Rewrite $f(t) = 100(1.21)^{t/2}$ in the form $f(t) = 100b^t$. What is b?

a. 1.1 ✓

b. 1.21

c. 1.4641

d. 0.605

Using exponent laws: $(1.21)^{t/2} = ((1.21)^{1/2})^t = (\sqrt{1.21})^t = (1.1)^t$. Thus $b = 1.1$.

8 If $P(t) = 250e^{0.08t}$, what is the equivalent discrete annual growth factor rounded to four decimal places?

a. 1.0833 ✓

b. 1.0800

c. 1.0408

d. 1.1664

$e^{0.08t} = (e^{0.08})^t$. Since $e^{0.08} \approx 1.083287$, the annual factor is approximately 1.0833.

9 A medication decreases in the bloodstream by 30% every 6 hours. By what factor does it decrease per hour?

a. $(0.70)^{1/6}$ ✓

b. $(0.30)^{1/6}$

c. $0.70/6$

d. $(0.70)^6$

A decrease of 30% means a 6-hour factor of $1 - 0.30 = 0.70$. The hourly factor b satisfies $b^6 = 0.70 \Rightarrow b = (0.70)^{1/6}$.

Section 4: Comparing Linear and Exponential Growth

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10 A linear function $L(x) = 50x + 100$ and an exponential function $E(x) = 2^x$ are compared for large $x > 0$. Which statement is true?

a. **Eventually $E(x) > L(x)$ for all sufficiently large x** ✓

b. Eventually $L(x) > E(x)$ because its initial rate of change is 50

c. They will intersect infinitely many times

d. $L(x)$ and $E(x)$ grow at the same rate asymptotically

Any exponential growth function with base $b > 1$ will eventually exceed any linear or polynomial function for sufficiently large values of x.

11 Which data set represents an exponential function rather than a linear function?

a. $x: 1, 2, 3, 4$ **and** $y: 6, 18, 54, 162$ ✓

b. $x: 1, 2, 3, 4$ and $y: 6, 12, 18, 24$

c. $x: 1, 2, 3, 4$ and $y: 6, 11, 18, 27$

d. $x: 1, 2, 3, 4$ and $y: 6, 8, 12, 20$

In an exponential function, consecutive y-values over equal intervals have a constant ratio: $\frac{18}{6} = 3$, $\frac{54}{18} = 3$, $\frac{162}{54} = 3$. This confirms exponential behavior.

12 What is the domain and range of the function $f(x) = 4 - 3(2)^x$?

a. **Domain:** $(-\infty, \infty)$**; Range:** $(-\infty, 4)$ ✓

b. Domain: $(0, \infty)$; Range: $(-\infty, 4)$

c. Domain: $(-\infty, \infty)$; Range: $(4, \infty)$

d. Domain: $(-\infty, 4)$; Range: $(-\infty, \infty)$

Exponential functions are defined for all real x, so domain is $(-\infty, \infty)$. Since $2^x > 0$ for all x, $-3(2)^x < 0$, which makes $4 - 3(2)^x < 4$. The range is $(-\infty, 4)$.

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